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Final Report

Origins of Mars's Moons: Physics-Structured ML Using Asteroid Tidal-Disruption SPH Simulations

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Abstract

This paper investigates asteroid tidal disruption during close encounters with Mars, a process in which tidal forces can fragment an incoming asteroid and leave part of the resulting debris gravitationally bound. Such retained material may be relevant to proposed formation pathways for Mars' moons: Phobos and Deimos. However, exploring the encounter parameter space with high-fidelity smoothed-particle hydrodynamics (SPH) simulations is computationally expensive. To support more efficient exploration, SPH particle outputs are post-processed into fragmentation diagnostics and bound mass fraction (BMF), and machine-learning surrogates are trained using leakage-safe grouped cross-validation. Physics-derived features, advanced boosting models, regime-aware approaches, and two-stage hurdle architectures are evaluated alongside tree-based baselines. The baseline Random Forest achieves out-of-fold $R^2=0.8971$, improving to $R^2=0.9225$ with physics-derived features. The best overall model, an NGBoost hurdle architecture, reaches $R^2=0.9485$ and $RMSE=0.0211$, while a CatBoost hurdle gives the lowest MAE of 0.0122. Controlled parameter slices and coverage-error diagnostics show that predictive reliability depends strongly on local SPH support. The resulting surrogate is therefore suitable for rapid in-domain screening and prioritisation, while sparse, edge, or physically detailed cases should still be evaluated with full SPH simulation.

1. Introduction

1.1 Research Problem and Questions

The origin of Mars's moons Phobos and Deimos remains unresolved. One proposed pathway is that a close asteroid encounter with Mars can produce tidal disruption: differential Martian gravity stretches and fragments the incoming body, after which some debris escapes and some remains gravitationally bound. That retained material is relevant to moon-formation hypotheses because it provides a possible source population from which later dynamical evolution and accretion could act (Kegerreis et al., 2024).

High-fidelity smoothed-particle hydrodynamics (SPH) is well suited to this problem because it follows moving material parcels through large deformation, fragmentation and gravitational interaction. However, each new setup is computationally expensive, making exhaustive exploration of asteroid mass, closest approach, encounter speed, spin and numerical choices impractical. The objective of this paper is therefore not to replace SPH, but to turn an existing simulation archive into a fast decision layer that can screen candidate encounters, identify promising retained-mass cases, and indicate when a full SPH calculation is still required.

The report addresses three research questions: (1) how do orbital and physical parameters influence disruption and bound retention; (2) which inputs most strongly control fragment formation and retained debris; and (3) can machine-learning models predict outcomes across the

sampled domain with explicit trust checks? The final output is a physics-structured surrogate and decision workflow rather than an unqualified black-box predictor.

1.2 Significance and Originality

The work is significant both scientifically and computationally. Scientifically, the archive probes a mechanism that could supply debris to the Martian system. Computationally, it addresses a recurring problem in simulation-led science: a detailed solver can answer one configuration well, but the cost of many configurations limits parameter-space exploration. A surrogate can map broad behaviour almost instantly once trained, allowing expensive simulations to be concentrated where they add the most value.

The main originality is the combination of prediction with explicit evidence about when prediction should be trusted. The workflow converts particle-level SPH outputs into run-level targets, evaluates models on entirely held-out physical simulations, re-expresses the inputs using physics-derived features, checks whether learned trends agree with SPH slices, measures local support and held-out error, and converts these diagnostics into an SPH recommendation. This makes the surrogate operationally useful without presenting machine learning as 'better physics' than SPH.

1.3 Literature Review and Modelling Rationale

Multiple scenarios have been proposed for Phobos and Deimos. Giant-impact calculations can generate debris disks around Mars (Citron et al., 2015; Canup & Salmon, 2018), transient inner moons may affect outer-disk accretion (Rosenblatt et al., 2016), and dynamical arguments have motivated a disrupted common progenitor (Bagheri et al., 2021). No single scenario yet connects debris production, capture, long-term orbital evolution and the final moons in a complete way.

Kegerreis et al. (2024) directly motivates this paper by demonstrating disruptive partial capture during close asteroid encounters with Mars. The simulations show that appreciable bound debris can arise across variations in periapsis, velocity, mass and spin, but a new configuration still requires a new simulation. This paper asks whether the archive can be learned as a structured dataset so that future configurations can be screened before SPH is launched.

SPH is a mesh-free Lagrangian method in which computational particles move with the represented material (Price, 2012). It is appropriate for violent deformation, but the derived fragment catalogue depends on post-processing. In this paper, friends-of-friends (FoF) groups nearby particles using a linking length; therefore fragment counts and fragment masses partly reflect a methodological choice as well as encounter physics (More et al., 2011). Bound mass is physically distinct: fragments are classified using their specific orbital energy relative to Mars.

The processed learning problem is small, tabular and nonlinear rather than a field-level surrogate. Tree ensembles are consequently a sensible baseline: Random Forest averages many

nonlinear trees, while Gradient Boosting corrects residual errors sequentially (Breiman, 2001; Friedman, 2001). This choice is also consistent with evidence that tree-based models remain strong on typical medium-sized tabular datasets (Grinsztajn et al., 2022). Physics-informed neural networks, by contrast, are most natural when governing-equation residuals can be enforced across continuous spatial or temporal domains (Raissi et al., 2019), which is not the form of the present run-level dataset.

Model structure was nevertheless kept as an empirical question. Beyond the tree baseline, this paper tests XGBoost and CatBoost boosting, a regime-aware mixture of experts, two-stage hurdle architectures, and a multi-output Gaussian Process. These alternatives test whether BMF is best represented as one smooth response, as regime-dependent behaviour, or as two linked questions: whether any material remains bound and, if so, how much.

2. Methodology

2.1 Data Archive, Inputs, and Post-Processing

The archive contains 489 SPH simulation outputs, totalling approximately 218 GB of HDF5 particle data. Each output records particle positions, velocities, mass, density, pressure and material type. The setup parameters describe the incoming asteroid and numerical configuration; the post-processing pipeline maps those particle snapshots to fragment- and bound-retention summaries suitable for exploratory analysis and machine learning.

Table 1. Main SPH setup and post-processing parameters.
Coverage is strongly uneven across the archive.

Parameter	Coverage / range	Role in the workflow
Asteroid mass	A_{1800} - A_{2100} ; A_{2000} dominates	Physical size/mass scale; larger bodies generally retain more bound debris.
Periapsis	r_{11} - r_{30} ; r_{12} and r_{16} common	Closest approach to Mars; smaller values give stronger tidal forcing.
Encounter speed	v_{00} - v_{30} ; v_{00} dominates	Higher speed makes capture less favourable.
Spin	none, x, y, z, mz	Rotation rate and orientation modify disruption geometry; coverage is uneven.
Resolution	n_{50} - n_{70} ; n_{65} dominates	Numerical resolution; non- n_{65} cases are sparse.

Parameter	Coverage / range	Role in the workflow
FoF linking length	0.0001-0.0126; 0.004 common	Post-processing grouping threshold; affects measured fragments but is not an encounter-physics driver.

FoF first partitions the particle debris into connected groups. For each group g , its mass is the sum of its particle masses, $M_g = \sum_i m_i$. The centre-of-mass position and velocity are then used to calculate the group’s specific orbital energy relative to Mars. A group is classified as bound when $\epsilon_g < 0$ and unbound when $\epsilon_g \geq 0$. BMF is the sum of bound fragment mass divided by the original parent-asteroid mass. The supporting targets include bound fragment count, largest/average bound fragment mass, total fragment count and largest-remnant quantities. BMF is a retention proxy: it indicates material available around Mars, not proof that a stable disk or moon ultimately forms.

2.2 Learning Tasks and Model Families

The primary learning task is regression of continuous BMF. This preserves distinctions between, for example, 2%, 9% and 25% retained mass. A supporting classification sanity check asks whether BMF exceeds an operational 10% threshold; Logistic Regression provides a transparent linear-probability baseline, with tree classifiers used as nonlinear comparators. The 10% value is a screening threshold, not a universal moon-formation criterion.

The baseline regressors are Random Forest and Gradient Boosting. The advanced single-stage models, XGBoost and CatBoost, test whether stronger regularised boosting improves difficult cases; CatBoost is also convenient for categorical inputs such as spin direction. A mixture of experts tests regime-aware modelling, allowing different sub-models to specialise in zero-retention, transition and high-retention regions. Hurdle models separate BMF into two stages: first zero versus positive retention, then the magnitude of positive BMF. Beta-, CatBoost- and NGBoost-based hurdle variants were tested. Finally, a multi-output Gaussian Process tests whether a smooth probabilistic representation is appropriate for the response.

2.3 Leakage-Safe Grouped Validation and Metrics

The archive contains repeated post-processing rows from the same physical SPH simulation, especially when different FoF linking lengths are applied. A random row-wise split would therefore leak near-duplicate physical cases between training and testing. Five-fold GroupKFold cross-validation was used instead: every row derived from one physical simulation file remains in the same fold. Each out-of-fold (OOF) prediction is consequently made by a model that did not train on that physical simulation.

Regression is assessed with R^2 , mean absolute error (MAE) and root mean squared error (RMSE). R^2 measures the fraction of variation in held-out BMF explained by the model; it is not

a percentage accuracy. MAE measures the average absolute error directly in BMF units, so MAE = 0.012 corresponds to about 1.2 percentage points of BMF. RMSE gives greater weight to occasional large mistakes. Classification uses balanced accuracy, F1 and ROC AUC. Global scores are necessary but not sufficient: later diagnostics test whether accuracy is locally supported and physically sensible.

2.4 Physics-Derived Features

The raw setup variables describe the experiment, but a small tree model should not be forced to rediscover every physical transformation from limited data. I therefore retained the raw inputs and added deterministic features computed only from those inputs. No bound-mass or fragment outcome is used in feature engineering, so these features are inference-safe.

Table 2. Physics-derived features evaluated in addition to the original setup variables.

Feature	Type	Physical motivation
v_{∞}^2	Simple transform	Energy-style speed proxy because kinetic terms scale with velocity squared.
$1/r_p$	Simple transform	Re-expresses periapsis so very close encounters have larger numerical weight.
$r_p \times v_{\infty}$	Simple interaction	Distinguishes close/slow, close/fast, distant/slow and distant/fast cases.
$1/P_{\text{spin}}$	Simple transform	Spin frequency, so faster rotation maps to a larger value.
Asteroid radius	Simple transform	Converts mass to a direct geometric size scale using the adopted density.
Encounter eccentricity	Composite physics feature	Combines periapsis and encounter speed to separate near-parabolic from hyperbolic fly-bys.
Time within $2 R_{\text{Mars}}$	Composite physics feature	Trajectory-duration proxy inside a strong-interaction region.
Time within tidal-disruption radius	Composite physics feature	Exposure time inside a Roche-like tidal threshold.

The value of these transforms was tested empirically rather than assumed. This distinction matters because a feature can sound physically meaningful yet fail to improve held-out prediction. The comparison therefore asks whether the same raw information has been expressed in a form that generalises better.

2.5 Trust and Decision Logic

A surrogate prediction is accepted only conditionally. The workflow checks five complementary properties: accuracy on unseen physical simulations; physically sensible response when one parameter is varied; support from nearby archive cases; stability across model families; and whether the case lies within the sampled joint domain. These checks support interpolation but explicitly lower trust for sparse, edge or extrapolative configurations.

The final decision also separates disruption from capture. Disruption severity is described by the largest-remnant fraction $f_{LR} = M_{\text{largest}}/M_{\text{parent}}$: $f_{LR} > 0.90$ is mostly intact, 0.50-0.90 indicates partial breakup, and $f_{LR} \leq 0.50$ indicates severe disruption. Retention is then assessed independently through BMF and the absolute Mars-bound mass $M_{\text{bound}} = \text{BMF} \times M_{\text{parent}}$. This two-stage screen reflects the fact that severe breakup can still leave almost no mass bound to Mars (Richardson et al., 1998; Walsh & Richardson, 2006).

3. Results

3.1 Exploratory Physical Trends

3.2 Eccentricity and Feature Engineering

3.3 Baseline and Advanced Model Performance

3.4 What Drives BMF Prediction?

3.5 Controlled Behaviour Checks

3.6 Coverage, Interpolation and Failure Modes

3.7 Decision-Support Interpretation

4. Discussion

5. Conclusions, Limitations and Future Work

5.1 Conclusions and Answers to the Research Questions

5.2 Limitations

5.3 Future Work

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Appendix A. Decision-Support Prototype

The interface translates the modelling results into a user-facing screening workflow. It accepts physical encounter inputs, shows disruption and bound-retention estimates, reports coverage/domain support, and returns either an ML-screening outcome or an SPH follow-up recommendation. The tool is distributed through the project repository and can be accessed by cloning the repo, running `pip install -e.` and launching the packaged local API/dashboard with `mars-flyby-dashboard`.

Figure A1. Mars Tidal Disruption Screening Explorer. The example reports strong fragmentation, predicted BMF, estimated Mars-bound mass, support diagnostics and an SPH recommendation.